

Exercise 3 – RFID

Dynamic-Frame ALOHA Collision Resolution Efficiency

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1 Introduction

The aim of this short report is to determine the expected length of the arbitration period in a Dynamic-Frame ALOHA RFID system with four tags ($N = 4$) and, from that, to compute the overall collision-resolution efficiency η for six different initial frame sizes $r_1 \in \{1, \dots, 6\}$. Finally, the optimum value of r_1 is identified.

2 Expected arbitration length $L_4(r_1)$

2.1 Recursive formulation

Let $L_n(r)$ be the expected number of slots required to identify n backlogged tags when the current frame length is r . With uniform slot selection we have

$$L_n(r) = r + \sum_{i=0}^{\min(r,n)} P_n(s = i; r) L_{n-i}(n-i) \quad (1)$$

with base cases $L_0 = 0$ and $L_1 = 1$. The exercise provides $L_2(2) = 4$ and $L_3(3) = \frac{51}{8}$.

2.2 Computed values for $N = 4$

Applying (1) for every r_1 gives

r_1	$P_4(s=0)$	$P_4(s=1)$	$P_4(s=2)$	$P_4(s=4)$	$L_4(r_1)$
1	1	0	0	0	≈ 9.824
2	$\frac{1}{2}$	$\frac{1}{2}$	0	0	≈ 9.600
3	$\frac{7}{27}$	$\frac{8}{27}$	$\frac{4}{9}$	0	≈ 8.956
4	$\frac{40}{256}$	$\frac{48}{256}$	$\frac{144}{256}$	$\frac{24}{256}$	≈ 8.824
5	$\frac{65}{625}$	$\frac{80}{625}$	$\frac{360}{625}$	$\frac{120}{625}$	≈ 9.038
6	$\frac{96}{1296}$	$\frac{120}{1296}$	$\frac{720}{1296}$	$\frac{360}{1296}$	≈ 9.466

3 Efficiency

The collision-resolution efficiency is

$$\eta = \frac{N}{L_4(r_1)} = \frac{4}{L_4(r_1)}.$$

Table 1 summarises the numerical results, while Figure 1 (right) provides a visual trend without interrupting the text flow.

4 Discussion

The highest efficiency ($\eta \approx 0.453$) is achieved when the initial frame equals the backlog ($r_1 = N = 4$). Shorter frames

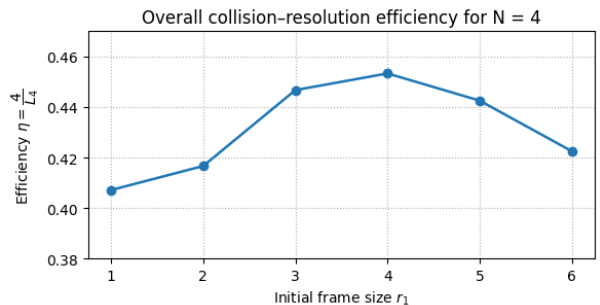


Figure 1: Efficiency η versus initial frame size r_1

Table 1: Efficiency versus initial frame size

r_1	$L_4(r_1)$ (slots)	η
1	9.824	0.407
2	9.600	0.417
3	8.956	0.447
4	8.824	0.453
5	9.038	0.443
6	9.466	0.423

cause excessive collisions, whereas longer frames waste idle slots—both lengthening the arbitration period.